# How Much Data Does an Information Coefficient Need?

## The Sampling Distribution of a Cross-Sectional IC, and Why Breadth Does Not Buy Precision

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**Abstract**

A cross-sectional information coefficient (IC) is usually treated as if the stocks in each period were an independent sample, so that $N$ stocks give a standard error of order $1/\sqrt{N}$. Any real signal is correlated with factor loadings, though, so each period's IC also picks up that period's factor return, a single draw shared by every stock. I show that $\mathrm{Var}(\widehat{IC}_t) = \sigma_F^2 + \kappa/N$ with $\kappa \approx 1$, where the floor $\sigma_F^2$ does not shrink as stocks are added, so effective sample size is capped at $1/\sigma_F^2$. Simulation confirms the formula. On 2,559 US stocks over 2000-02–2026-02, the IC volatility of $12-1$ momentum levels off at $\sigma_F = 0.1397$: a cross-section of 1,552 names is worth about 50 independent observations, while a randomly permuted signal follows the classical formula. Detecting a true IC of 0.03 at 80% power takes 14.5 years of monthly data, or 27.2 years at the $t > 3$ hurdle, and no ten-year window is long enough. The floor itself is the strategy risk of Qian and Hua (2004); what this paper adds is what it means for inference.

**Keywords:** information coefficient, statistical power, effective sample size, fundamental law of active management, factor models, replication
**JEL classification:** C12, C15, G11, G12

## 1 Summary in plain language

An investor who thinks they can predict which stocks will do well might test it like this. Each month, score every stock, wait a month, and check whether the high-scoring stocks actually did better. The correlation between the scores and what happened is called the information coefficient. Averaged over years of monthly data, published figures are often around 0.03. That is a weak relationship (a perfect forecast would score 1), but applied to thousands of stocks every month it is enough to run a business on.

How sure can anyone be that a measured 0.03 is real and not luck? The obvious answer is: very sure. Two thousand stocks a month for twenty-five years is about half a million observations. This paper shows why that answer is wrong and works out by how much.

Stocks do not move independently. In any month a few big shared movements push every stock at once: the whole market rises, say, or cheap stocks beat expensive ones. And any signal a person would actually use is related to which stocks are exposed to those movements. Recent winners tend to be the jumpier stocks; cheap stocks tend to be the riskier ones. So part of each month's measured correlation just reflects which way the market happened to lean that month.

Adding more stocks does not fix this, because there is only one market movement per month and it touches everything. Noise that is specific to individual companies washes out as names are added; the shared part does not.

I work out how large the shared part is, check the formula on simulated data where the right answer is known, and then measure it on twenty-five years of US stock prices. A cross-section of about 1,552 stocks turns out to carry as much information as roughly 50 independent observations.

Showing that a true information coefficient of 0.03 is different from zero takes about 14.5 years of monthly data, or about 27.2 years under the stricter standard the field now uses because so many strategies have been tried. Over any ten-year stretch the smallest effect that can be reliably detected is 0.0361, which is larger than 0.03. A study that reports such an effect "holds in every decade" is claiming more than its data can show.

Momentum (buying stocks that have risen over the past year) gives an average IC of 0.0123 over 300 months in my data, with a $t$-statistic of 1.51, which is not distinguishable from zero.

## 2 Motivation

The information coefficient is the standard summary statistic for a return-predicting signal, in academic work and in industry. It is reported constantly and its precision is reported rarely. When a standard error does accompany it, the usual construction is Fama–MacBeth: treat the $T$ per-period ICs as a sample and use $t = \sqrt{T}\,\overline{IC}/\hat{\sigma}_{IC}$ (Fama and MacBeth, 1973). The procedure is sound; the problem is the intuition about how much precision the cross-section supplies.

For $N$ independent pairs, the sample correlation has a standard error of about $1/\sqrt{N-1}$. With $N = 2000$ that is 0.022; averaged over $T = 300$ months it becomes 0.0013, which puts an IC of 0.03 about 23 standard errors from zero. Under this arithmetic no cross-sectional IC of plausible magnitude could ever fail to be significant, and questions of statistical power never arise.

Several facts do not fit this picture. Realised IC series have standard deviations around 0.10–0.15, five or six times the classical figure. Practitioners have long treated the volatility of the IC as a risk in its own right (Qian and Hua, 2004). And the empirical anomaly literature has the kind of replication problem that low power produces: Hou et al. (2020) find that 65% of 452 anomalies fail to clear $|t| > 1.96$ once microcaps are handled, rising to 82% at a multiple-testing hurdle, and that surviving effects are typically much smaller than first reported. McLean and Pontiff (2016) document returns 26% lower out of sample and 58% lower after publication.

I derive the sampling distribution of the cross-sectional IC under a factor model, check it by simulation, and measure its parameters on real data. The result turns the vague sense that ICs are noisy into a sample-size requirement that any particular study can be checked against. Section 3 sets out which parts of this are already known.

## 3 Related work

**The fundamental law, and what it already contains.** Grinold (1989) gave the fundamental law of active management, $IR \approx IC \cdot \sqrt{BR}$, with $BR$ the number of independent bets. The law is about attainable performance, not about estimation, but its breadth term is the source of the intuition this paper contradicts. Clarke et al. (2002) added the transfer coefficient to account for the information lost to portfolio constraints.

Two papers come closest. Qian and Hua (2004) showed that the volatility of the IC through time is a separate, non-diversifiable component of active risk, which they named *strategy risk*, and proposed the ratio of mean IC to IC standard deviation as the right performance statistic. Those are the same two moments whose sampling behaviour I study. Ding and Martin (2017) then derived a fundamental law in which the information ratio is increasing in the mean of $IC_t$, decreasing in its volatility, and subject to an absolute upper bound as $N \to \infty$.

The existence of a non-diversifiable component of IC variance is therefore not my finding; it is in both papers. What they do not cover is the inference side: the sampling distribution of $\overline{IC}$, an effective sample size for a cross-section, and the number of periods needed before a claimed IC can be distinguished from zero. Ding and Martin answer a portfolio-construction question (how good can a manager get?), and I answer an inference question (how much data does a claim need?), using the same variance term.

**What sets precision in cross-sectional tests.** Ang et al. (2020) show that the precision of cross-sectional risk-premium estimates depends on the cross-sectional dispersion of factor loadings: grouping stocks into portfolios estimates each beta more precisely but shrinks that dispersion, and the standard errors of the risk premia get larger. Lewellen et al. (2010) make a related point about the apparent success of asset pricing models on portfolios with strong factor structure. My setting is related. Here too the precision of a cross-sectional estimate depends on the structure of the cross-section, and counting assets overstates it once a common component is present.

**Multiple testing and replication.** Harvey et al. (2016) argue that the sheer number of factors tested requires a hurdle of about $t > 3.0$ rather than 1.96. Hou et al. (2020) document the resulting replication failures, and Chen and Zimmermann (2022) provide an open catalogue of 319 published predictors, which fixes the order of magnitude of the multiple-testing problem. Chen (2021) argues that $p$-hacking alone cannot account for the largest published $t$-statistics. This literature asks whether a given significance threshold is strict enough. I ask an earlier question, whether a study has the power to reach any threshold at all. The two interact: a stricter hurdle raises the required sample.

**Power and magnitude errors.** Outside finance, Ioannidis (2005) argued that low power makes a large share of published positive findings false, and Gelman and Carlin (2014) showed that it also inflates the published magnitudes of effects that are real. I use Gelman and Carlin's exaggeration ratio in §5.7.

# 4 Setup

Fix a rebalancing date $t$ with $N$ assets. Let $s_{i,t}$ be the signal, standardised cross-sectionally so that $\mathrm{mean}_i(s) = 0$ and $\mathrm{var}_i(s) = 1$, and let $r_{i,t+1}$ be the return earned over the following period. The realised IC is the cross-sectional correlation

$$\widehat{IC}_t \;=\; \frac{\mathrm{cov}_i(s_{\cdot,t},\, r_{\cdot,t+1})}{\mathrm{sd}_i(r_{\cdot,t+1})}, \tag{1}$$

and the reported statistic is $\overline{IC} = T^{-1}\sum_{t=1}^{T} \widehat{IC}_t$. Throughout, $\mathrm{mean}_i$, $\mathrm{cov}_i$ and $\mathrm{sd}_i$ denote moments taken across assets at a fixed date; expectations and variances without a subscript are over repeated draws of the return process.

Returns follow a one-factor model,

$$r_{i,t+1} \;=\; \lambda s_{i,t} \;+\; \beta_i f_{t+1} \;+\; \varepsilon_{i,t+1}, \tag{2}$$

with $f_{t+1}$ the factor return, $\mathrm{var}(f) = \sigma_f^2$; $\varepsilon$ cross-sectionally uncorrelated with variance $\sigma_\varepsilon^2$ and independent of $f$; and $\lambda \geq 0$ the strength of genuine predictability. Write $v_\beta = \mathrm{var}_i(\beta)$ and

$$q \;\equiv\; \mathrm{cov}_i(s_{\cdot,t},\, \beta_\cdot), \tag{3}$$

the cross-sectional covariance between the signal and the factor loadings. Because $s$ is standardised, $q$ is the correlation between signal and loading times $\mathrm{sd}_i(\beta)$.

**Assumptions, and where they are used.** *(A1) One factor.* Used only for notational economy; Appendix B gives the multi-factor statement, in which $q$ becomes a vector and $q^2\sigma_f^2$ becomes $q'\Sigma_f q$. Nothing qualitative changes. *(A2) $q \neq 0$.* This is the key assumption, and it separates a real signal from a random one. Any signal built from prices, valuations or accounting data is correlated with risk exposures across stocks. §8 builds a signal with $q = 0$ and shows that it follows the classical formula, so the assumption matters. *(A3) $\varepsilon$ cross-sectionally uncorrelated.* Residual correlation (industry clustering) would add a further non-diversifiable term, so this assumption makes my result conservative. *(A4) $q$ constant through time.* Relaxed in §5.5; time variation in $q$ or in $\lambda$ adds to the variance of $\widehat{IC}_t$ without biasing $\overline{IC}$. *(A5) Pearson correlation.* The derivation is for (1). In practice I use Spearman, which is Pearson applied to ranks, so the algebra carries over with $s$ and $r$ replaced by their rank transforms; §8 reports both. *(A6) Normality.* The proof treats each asset's $(s_i, \beta_i, \varepsilon_i)$ as an independent draw from one joint normal distribution, which is what lets the classical formula apply once $f$ is fixed. Ranks make the empirical work less sensitive to this.

# 5 The sampling distribution

## 5.1 The classical benchmark

If $(s_i, r_i)$ were $N$ independent draws from a bivariate normal with correlation $\rho$, the sample correlation would satisfy, to first order,

$$\mathrm{Var}(\widehat{IC}_t) \;=\; \frac{(1-\rho^2)^2}{N-1} \;\approx\; \frac{1}{N} \qquad \text{for small } \rho, \tag{4}$$

which is also the variance of the Spearman coefficient under independence, and the basis of Fisher's variance-stabilising transform (Fisher, 1921). Equation (4) is correct, but it assumes independent pairs, and a cross-section of stock returns does not supply them.

## 5.2 Decomposition

Substitute (2) into the numerator of (1). Cross-sectional covariance is linear in its second argument, and $\mathrm{var}_i(s) = 1$, so

$$\mathrm{cov}_i(s, r_{t+1}) = \lambda \underbrace{\mathrm{var}_i(s)}_{=1} + f_{t+1} \underbrace{\mathrm{cov}_i(s, \beta)}_{=q} + \underbrace{\mathrm{cov}_i(s, \varepsilon_{t+1})}_{\equiv u_t} = \lambda + q f_{t+1} + u_t. \tag{5}$$

The first term is the signal, the second is the factor leak, and the third is idiosyncratic noise. Since $\varepsilon$ is cross-sectionally uncorrelated with variance $\sigma_\varepsilon^2$ and $s$ is standardised,

$$\mathrm{E}[u_t] = 0, \qquad \mathrm{Var}(u_t) = \frac{\sigma_\varepsilon^2}{N}. \tag{6}$$

The denominator of (1) converges by the law of large numbers, for fixed $f_{t+1}$, to

$$\mathrm{sd}_i(r_{t+1}) \xrightarrow{N\to\infty} \sqrt{\sigma_\varepsilon^2 + v_\beta f_{t+1}^2 + O(\lambda)}. \tag{7}$$

So

$$\widehat{IC}_t \;=\; \underbrace{\frac{\lambda}{\mathrm{sd}_i(r_{t+1})}}_{\text{signal}} \;+\; \underbrace{\frac{q f_{t+1}}{\mathrm{sd}_i(r_{t+1})}}_{\text{factor leak: } O(1)} \;+\; \underbrace{\frac{u_t}{\mathrm{sd}_i(r_{t+1})}}_{O(N^{-1/2})}. \tag{8}$$

Only the third term shrinks with breadth. The second does not depend on $N$ at all: it is one random number, $f_{t+1}$, scaled by a fixed constant $q$. Averaging across many stocks removes what is specific to each stock but leaves what they have in common, and a signal correlated with factor loadings turns that common shock into apparent skill or apparent failure.

### 5.3 The variance floor

**Proposition 1** (Variance of the cross-sectional IC)**.** *Under* (2)*, (A1)–(A3) and (A6),*

$$\mathrm{Var}(\widehat{IC}_t) \;=\; \sigma_F^2 \;+\; \frac{\kappa}{N} \;+\; O(N^{-3/2}), \tag{9}$$

*where the coefficient on* $1/N$ *is*

$$\kappa \;=\; \mathrm{E}_f\Big[\big(1-g(f)^2\big)^2\Big] \;-\; \mathrm{Cov}_f\big(g(f),\, g(f)\,(1-g(f)^2)\big) \;=\; 1 - 2\rho^2 - 3\sigma_F^2 + O(g^4), \tag{10}$$

*with* $\rho = \mathrm{E}[\widehat{IC}_t]$*, and the floor is*

$$\sigma_F^2 \;=\; \mathrm{Var}_f[\,g(f)\,], \qquad g(f) \;=\; \frac{\lambda + qf}{\sqrt{\sigma_\varepsilon^2 + v_\beta f^2 + \lambda^2 + 2\lambda q f}}. \tag{11}$$

The proof is the decomposition above plus the law of total variance, conditioning on $f_{t+1}$; Appendix A gives it in full.

The floor in (11) has two properties used later.

First, $g$ is a *nonlinear* function of $f$: the factor appears in the numerator and, through the cross-sectional dispersion of returns, in the denominator. The floor is the variance of that function, not a ratio of variances. Taking the ratio instead gives the tempting closed form

$$\sigma_F^2 \;\approx\; \frac{q^2\sigma_f^2}{\sigma_\varepsilon^2 + v_\beta \sigma_f^2} \qquad \text{or, more crudely,} \qquad \frac{q^2\sigma_f^2}{\sigma_\varepsilon^2}, \tag{12}$$

and §6 shows these are wrong by 14.9% and 25.2% respectively, so the code uses the exact form.

Second, $g$ is bounded: as $|f| \to \infty$, $g(f) \to \pm q/\sqrt{v_\beta}$. The model's IC distribution therefore has thinner tails than a normal, a point I return to in §10 because real IC series do not share that property.

### 5.4 Effective breadth

Define the effective number of observations as the number of independent draws that would produce the same per-period variance:

$$N_{\text{eff}} \;\equiv\; \frac{1}{\mathrm{Var}(\widehat{IC}_t)} \;=\; \frac{N}{1 + N\sigma_F^2} \;\xrightarrow{N\to\infty}\; \frac{1}{\sigma_F^2}. \tag{13}$$

Breadth therefore has a ceiling. Doubling the universe from 1,000 to 2,000 names raises $N_{\text{eff}}$ from $1/(\sigma_F^2 + 10^{-3})$ to $1/(\sigma_F^2 + 5\times10^{-4})$, which at the measured $\sigma_F = 0.1397$ moves it from 48.7 to 50.0: about one extra effective observation for a thousand extra stocks. $N_{\text{eff}}$ reaches half its ceiling at $N = 1/\sigma_F^2$, and beyond that point extra names add very little.

### 5.5 Aggregating over time

Averaging $T$ periods,

$$\operatorname{Var}(\overline{IC}) \;=\; \frac{\sigma_{IC}^2}{T}\left(1 + 2\sum_{k=1}^{K}\left(1 - \tfrac{k}{K+1}\right)\varrho_k\right), \tag{14}$$

where $\sigma_{IC}^2 = \operatorname{Var}(\widehat{IC}_t)$ and $\varrho_k$ is the order-$k$ autocorrelation of the IC series, handled by the Newey and West (1987) correction. If the true IC varies through time, with $\rho_t = \rho + \eta_t$ and $\operatorname{Var}(\eta) = \tau^2$, then $\sigma_{IC}^2$ picks up an extra $\tau^2$, which raises the variance without biasing $\overline{IC}$. Relaxing (A4) therefore makes the sample-size requirements below larger, never smaller. Together with (13), this means that past a modest $N$ the only way to gain precision is a longer sample.

### 5.6 Required sample size

Testing $H_0 \colon \rho = 0$ against a true IC of $\rho_1$, two-sided at level $\alpha$ with power $\pi$, the standard normal-approximation calculation gives

$$\boxed{T^* \;=\; \frac{\left(z_{1-\alpha/2} + z_{\pi}\right)^2 \sigma_{IC}^2}{\rho_1^2}} \tag{15}$$

and, inverted, the minimum detectable IC for a sample of $T$ periods,

$$\rho_{\min}(T) \;=\; \frac{\left(z_{1-\alpha/2} + z_{\pi}\right)\sigma_{IC}}{\sqrt{T}}. \tag{16}$$

Combining (15) with (13) gives a compact statement of the result. Since $\sigma_{IC}^2 = 1/N_{\text{eff}}$, the total number of *effective* observations a study commands is $T \cdot N_{\text{eff}}$, and the requirement is

$$T \cdot N_{\text{eff}} \;\geq\; \frac{\left(z_{1-\alpha/2} + z_{\pi}\right)^2}{\rho_1^2}. \tag{17}$$

For $\rho_1 = 0.03$ at 80% power and 5% significance, the right-hand side is 8,721. A monthly cross-section supplies at most $1/\sigma_F^2$ of them per period, no matter how large the universe.

### 5.7 Exaggeration of published estimates

If an estimate is reported only when significant, the reported magnitudes are drawn from the tails. Following Gelman and Carlin (2014), with $\hat{\rho} \sim N(\rho, \sigma^2)$ and $d = \rho/\sigma$, the exaggeration ratio is

$$M(d, z) \;=\; \frac{d\,\Phi(d - z) + \phi(z - d) - d\,\Phi(-z - d) + \phi(z + d)}{d\,\left[\Phi(d - z) + \Phi(-z - d)\right]}, \tag{18}$$

which is always above 1, grows quickly once power falls below about 50%, and approaches 1 as power approaches 1. Section 9 evaluates it at the IC volatility I measure.

## 6 Numerical verification

I check the derivation on simulated data, where the truth is known. I simulate (2) directly with $\gamma = 0.35$ (the correlation between signal and loading), $\operatorname{sd}_i(\beta) = 0.5$, $\sigma_f = 0.06$, $\sigma_\varepsilon = 0.10$, and $\lambda$ set so that the model's true mean IC is 0.0296, the magnitude in the motivating example. Cross-sections are drawn independently across periods, so that the aggregation results below test (15) in isolation from any autocorrelation question.

**Variance against breadth.** Table 1 and Figure 1 compare simulated $\mathrm{Var}(\widehat{IC}_t)$ against (9) across a 256-fold range of $N$.

| $N$ | simulated | predicted | error |
|---|---|---|---|
| 50 | 0.029022 | 0.028325 | −2.4% |
| 100 | 0.018320 | 0.018605 | +1.6% |
| 200 | 0.013642 | 0.013745 | +0.8% |
| 400 | 0.011296 | 0.011314 | +0.2% |
| 800 | 0.010010 | 0.010099 | +0.9% |
| 1600 | 0.009510 | 0.009492 | −0.2% |
| 3200 | 0.009208 | 0.009188 | −0.2% |
| 12800 | 0.009107 | 0.008960 | −1.6% |

Table 1: Simulated versus predicted variance of the per-period IC. Predicted values are $\sigma_F^2 + \kappa/N$ with $\sigma_F^2 = 0.008884$ from (11) and $\kappa = 0.972$ from (10). Fitting $V(N) = a + c/N$ to the simulated column returns $a = 0.008807 \pm 0.000063$ and $c = 0.997 \pm 0.014$, against a theoretical $c = \kappa = 0.972$.

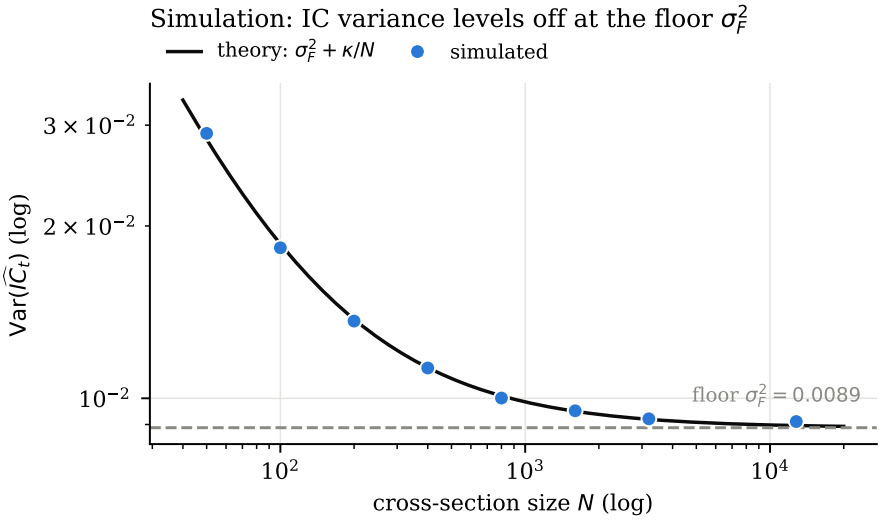



Figure 1: Simulated IC variance against cross-section size, with the analytic curve (9) and the floor (11). Variance falls as $1/N$ while the idiosyncratic term dominates, then flattens.

The fitted coefficient on $1/N$ is $0.997 \pm 0.014$ where theory predicts $\kappa = 0.972$, a gap inside two Monte Carlo standard errors. The fit is unweighted and so leans on the smallest $N$, where the $O(N^{-3/2})$ remainder still pushes $c$ up.

**Which expression for the floor.** The fitted intercept, $0.008807 \pm 0.000063$, is the simulated floor, and Table 2 compares it with the exact form (11) and the two closed forms (12). The exact form is 0.9% away, within Monte Carlo error. The closed form that keeps $v_\beta \sigma_f^2$ in the denominator is 14.9% too high, and dropping that term makes it worse.

**Coverage and power.** For Table 3 I draw 2,000 samples of $T$ periods each at $N = 400$. In each sample I record whether the Fama–MacBeth interval covers the true mean IC and whether the test rejects $\rho = 0$. From $T = 60$ on, coverage is within a percentage point of

| estimator | value | error | Monte Carlo $\sigma$ |
|---|---|---|---|
| exact leading order, (11) | 0.008884 | +0.9% | 1.2 |
| closed form $q^2\sigma_f^2/(\sigma_\varepsilon^2 + v_\beta\sigma_f^2)$ | 0.010115 | +14.9% | 20.7 |
| naive ratio $q^2\sigma_f^2/\sigma_\varepsilon^2$ | 0.011025 | +25.2% | 35.1 |

Table 2: Candidate expressions for $\sigma_F^2$ against the simulated floor $0.008807 \pm 0.000063$. Only the exact leading-order expression is within two Monte Carlo standard errors of it.

95% and simulated power is within about 0.015 of the analytic value, which supports using (15) for the sample-size calculations in §9. At $T = 24$ both are off by about three Monte Carlo standard errors, for the reason given in the caption.

| $T$ | coverage of 95% interval | power, simulated | power, analytic |
|---|---|---|---|
| 24 | 0.936 | 0.308 | 0.275 |
| 60 | 0.946 | 0.585 | 0.576 |
| 120 | 0.951 | 0.876 | 0.861 |
| 300 | 0.952 | 0.999 | 0.998 |

Table 3: Coverage and power at $N = 400$, true mean IC 0.0296. At $T = 24$ the simulated power is a little above the analytic value, and coverage a little below 95%, because the test uses a normal critical value where a $t$ distribution with few degrees of freedom would be more accurate.

# 7 Data

The stock list is the holdings of the iShares Russell 3000 ETF (IWV) on the day the data were downloaded, with sectors taken from the same holdings file. Daily prices for those stocks come from Yahoo Finance, downloaded with the open-source `yfinance` library. The result is one file (`russell3000_data.csv`) with daily open, high, low, close, volume and sector for 2,559 US tickers, 11,262,530 rows spanning 2000-01-03 to 2026-03-11. Closes are back-adjusted for splits and dividends, so successive ratios are total returns. I take the last observed close in each calendar month, form monthly returns, and discard the final partial month, giving 313 months from 2000-02 to 2026-02. The momentum sample has a median of 1,552 names per month.

Two signals are constructed, both from prices alone: $12 - 1$ momentum, the cumulative log return over the eleven months ending one month before formation (Jegadeesh and Titman, 1993), and one-month reversal, the previous month's return negated (Jegadeesh, 1990). Signals are measured at the end of month $t$ and matched to the return earned during month $t + 1$.

**Known defects.** The data has three known problems, and I checked each one directly (Appendix C). *(i) The universe is completely survivorship-biased.* All 2,559 tickers survive to the final date, with no delistings. This follows from how the file was built: the stock list is the index's membership on the download date, so any company that left the index earlier, through bankruptcy, a merger or deletion, is not in the data. *(ii) Some columns are transformed despite their names.* In an accompanying fundamentals file, a column named `book_to_market` is 83.2% negative and its exponential has median 0.447: it is log book-to-market. A column named `div_yield` takes only the values $\{0, 1, 2, 3\}$. I use neither file;

this check is one reason the analysis uses prices only. *(iii) The price snapshot cannot be rebuilt exactly*, since every later dividend rescales the back-history. Testing this would mean downloading the data again, which I did not do, so my original download is the input of record. I also checked that every close lies inside that day's high-low range, which suggests that closes and the other prices are adjusted the same way.

Section 10 discusses what defect (i) means for the results.

# 8 Empirical results

## 8.1 The realised IC series

| signal | $T$ | mean | sd | $t$ | $t_{NW}$ | AR(1) | median $N$ | $N_{\text{eff}}$ |
|---|---|---|---|---|---|---|---|---|
| $12-1$ momentum | 300 | 0.0123 | 0.1411 | 1.51 | 1.60 | -0.02 | 1,552 | 50 |
| 1-month reversal | 312 | 0.0221 | 0.1164 | 3.36 | 3.56 | -0.06 | 1,570 | 74 |

Table 4: Monthly Spearman IC, 2000-02 to 2026-02. $t$ is the Fama–MacBeth statistic $\sqrt{T}\,\overline{IC}/\hat{\sigma}_{IC}$; $t_{NW}$ applies a Newey–West correction with six lags.

Momentum's mean IC of 0.0123 carries $t = 1.51$: not distinguishable from zero. Under the classical formula it would have been overwhelming, since $0.0254/\sqrt{300} = 0.0015$ implies $t \approx 8$. Reversal's 0.0221 gives $t = 3.36$, which survives even the $t > 3$ hurdle of Harvey et al. (2016). Autocorrelation is negligible in both ($-0.02$ and $-0.06$), so the Newey–West correction moves the statistics only slightly, and upward: 1.60 and 3.56.

## 8.2 IC volatility and breadth

Proposition 1 makes a prediction that can be tested on real data without specifying any factor model. Draw random subsets of $N$ names, recompute the IC series, and see whether its variance traces $\sigma_F^2 + c/N$ with $c = 1$. Only 26% of months have 2,000 names, all of them late in the sample, so the subsamples stop at $N = 1000$. The thinnest month has 1,083 names, so every month enters at every size.

| $N$ | $12-1$ momentum | 1-month reversal | random signal | classical $1/\sqrt{N-1}$ |
|---|---|---|---|---|
| 50 | 0.2021 | 0.1866 | 0.1428 | 0.1429 |
| 100 | 0.1778 | 0.1548 | 0.1025 | 0.1005 |
| 200 | 0.1565 | 0.1354 | 0.0715 | 0.0709 |
| 400 | 0.1485 | 0.1255 | 0.0514 | 0.0501 |
| 800 | 0.1441 | 0.1191 | 0.0358 | 0.0354 |
| 1000 | 0.1436 | 0.1185 | 0.0311 | 0.0316 |
| fitted $\sigma_F$ | 0.1397 | 0.1136 | $\approx 0$ | — |
| fitted $c$ | 1.090 | 1.097 | 1.023 | — |

Table 5: Standard deviation of the realised IC series by cross-section size, averaged over five independent draws. The random signal is a per-period random permutation: it has $q = 0$ by construction. Its fitted floor is 0.000042, essentially zero, against momentum's 0.019514.

The fitted coefficient on $1/N$ is 1.090 for momentum and 1.097 for reversal, against a predicted 1, so the functional form fits. The overshoot of about 9% is larger than the random-signal control's (1.023 on the same months). A likely reason is assumption (A3): residual

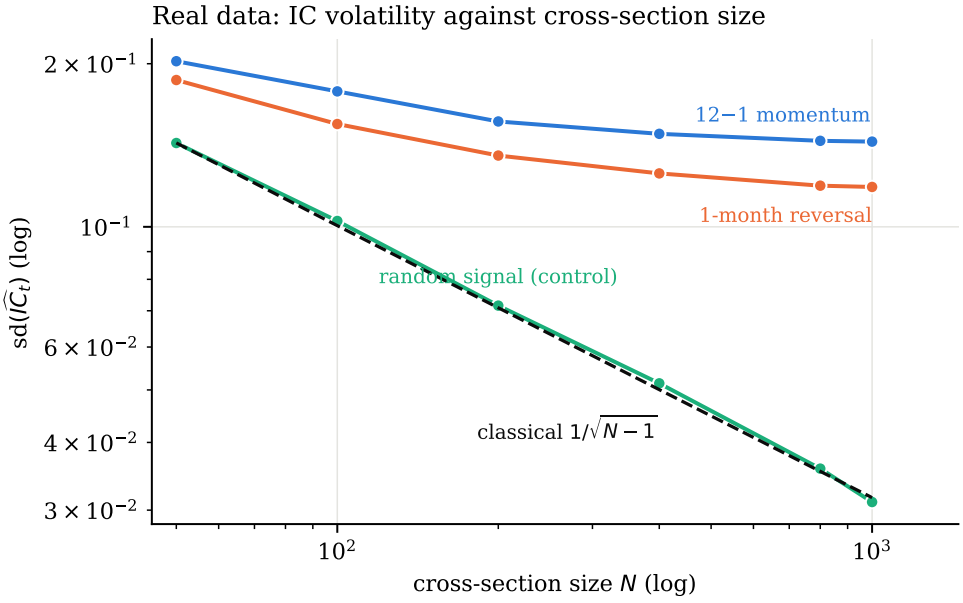



Figure 2: IC volatility against cross-section size on real data. The random-signal control lies on the classical curve throughout; both real signals flatten well above it.

returns are correlated within industries, so the diversifiable term shrinks a little more slowly than $1/N$. That changes the constant on $1/N$, not the floor.

The random-signal control follows $1/\sqrt{N-1}$ to within 3% over a twenty-fold range of $N$, with a fitted floor of 0.000042. If dependence between returns were enough to create the floor, a structureless signal would show one too, and it does not. The floor needs $q \neq 0$, a signal correlated with loadings, as (8) says. Because the control goes through the same code as the real signals, it also checks the IC estimator.

Momentum's IC volatility levels off at $\sigma_F = 0.1397$, so by (13) the ceiling on effective breadth is 51 observations. The cross-section actually used has 1,552 names and delivers 50.

### 8.3 A null with realistic dependence

A second estimate of the floor keeps both cross-sections intact and breaks only the timing: each month's signal is paired with a *different* month's returns. This gives the distribution of the IC when there is no predictive relation at all but the real factor structure is still there. A random permutation also creates a few pairs that are not nulls: a month matched with itself, or with a return already inside the signal's twelve-month look-back (for reversal these score exactly $-1$). Those pairs are dropped. Momentum's placebo has standard deviation 0.0766, against 0.0254 from the classical formula, so it is about 3.0 times wider even though predictability is exactly zero.

This placebo is a *lower* bound on $\sigma_F$. Misaligning months also severs any link between a signal's factor exposure in one month and factor volatility in the next, and if those covary the placebo understates the floor. That is why I use the subsampling curve as the main estimate and the placebo only as a check.

### 8.4 Robustness

**Rank versus raw correlation.** My derivation is for Pearson correlation; I report Spearman. Computed both ways on raw returns, momentum's IC standard deviation is 0.1411

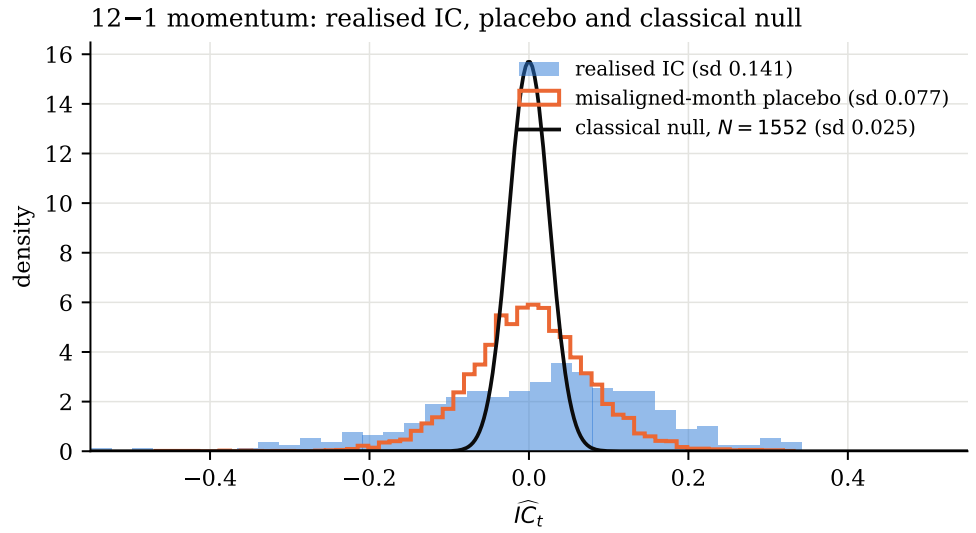



Figure 3: Three distributions of the same statistic: the classical null, the misaligned-month placebo, and the realised IC.

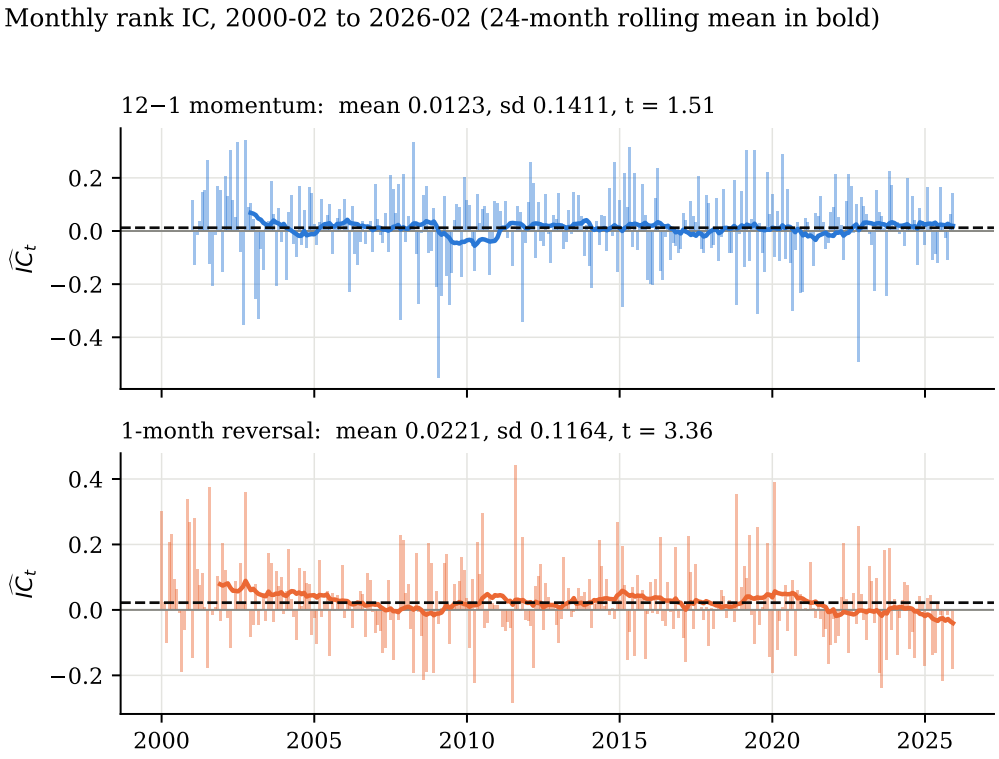



Figure 4: The two realised monthly IC series with 24-month rolling means. The paper uses their dispersion, not their level.

(Spearman) against 0.1410 (Pearson), effectively identical, so the main result does not depend on the rank transform. The *mean*, however, moves from 0.0123 to −0.0080 and changes sign. The panel contains a monthly return of +30,991%, and a point estimate that a single observation can flip is not one to build on, which is why I use ranks.

**Survivorship.** Restricting the sample to the 1,083 tickers with closes at the end of both January and February 2000 (the first monthly return), a cohort with a different survivorship pattern, moves the mean IC from 0.0123 to 0.0040 (about a third of its value), while the standard deviation moves from 0.1411 to 0.1539, about 9%. The paper relies on the second moment being stable, and here it is much more stable than the first.

# 9 Implications

**How much data a claimed IC of 0.03 needs.** At the measured $\sigma_{IC} = 0.141$, equation (15) at 80% power gives:

| hurdle | required $T$ (months) | years |
|---|---|---|
| $t > 1.96$, single test | 174 | 14.5 |
| $t > 3.00$ (Harvey et al., 2016) | 326 | 27.2 |
| $t > 3.76$, Bonferroni over 300 anomalies | 468 | 39.0 |

Table 6: Sample length required to distinguish a true IC of 0.03 from zero.

A study using the CRSP sample from 1963 to about 2020 ($\approx 57$ years) clears the single-test bar comfortably and the multiple-testing bar with room. A study on twenty years of one country's data, or on a subperiod, or on any market with a short history, does not.

**Decade-by-decade claims.** Equation (16) gives the smallest IC detectable at 80% power:

| sample length | 5 yr | 10 yr | 20 yr | 25 yr | 57 yr |
|---|---|---|---|---|---|
| $\rho_{\min}$ | 0.0510 | 0.0361 | 0.0255 | 0.0228 | 0.0151 |

Over ten years the minimum detectable IC is 0.0361, larger than 0.03, so a paper that reports a mean IC of 0.03 *and* claims the effect holds in every decade is claiming something its decade-long windows cannot establish at conventional power. Either the subperiod results are underpowered and their significance is partly luck, or the true IC is larger than 0.03. Subperiod robustness tables are common, and this objection can be checked against any of them.

**Effective size of the panel.** The panel contains 465,750 stock-months. By (17) it carries 15,078 effective observations, about 31 times fewer, against the 8,721 needed to detect an IC of 0.03 at 80% power. Twenty-five years of a broad universe clears that bar by less than a factor of two.

**Exaggeration.** Evaluating (18) at $\sigma_{IC} = 0.141$ for a true IC of 0.03:

| sample length | 5 yr | 10 yr | 20 yr | 25 yr | 57 yr |
|---|---|---|---|---|---|
| exaggeration ratio | 1.61 | 1.25 | 1.05 | 1.03 | 1.00 |

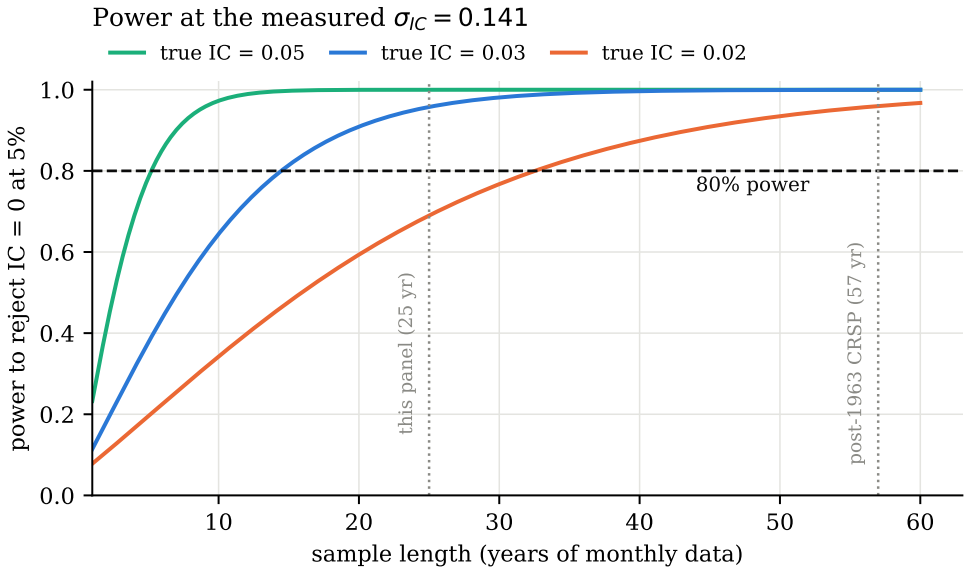



Figure 5: Power to reject $IC = 0$ against sample length, at the measured $\sigma_{IC} = 0.141$.

The winner's curse here is a short-sample problem. At five years a published IC of this size is inflated by about 1.61×, and by twenty-five years the effect is negligible. The result therefore supports the claim that low power inflates estimates from short samples and subperiods, but not long-sample anomaly estimates in general.

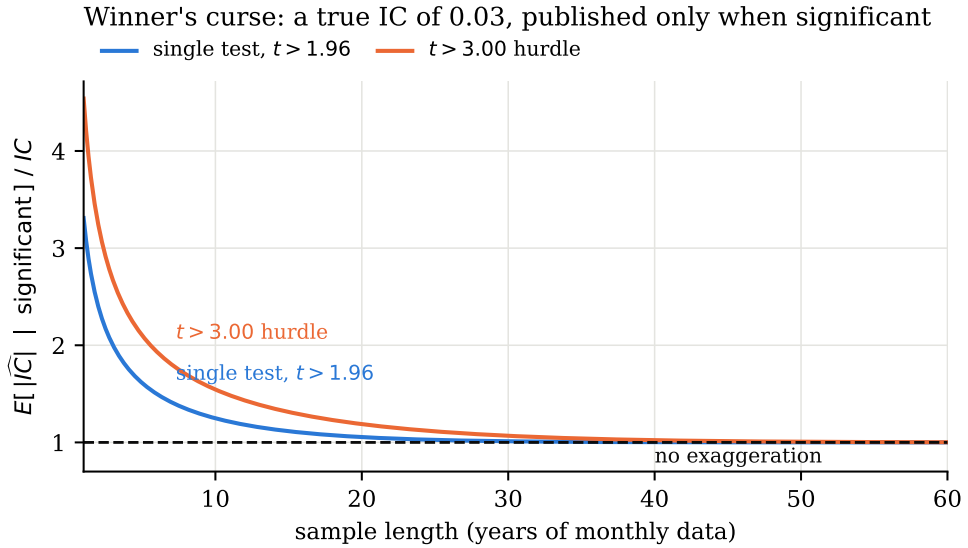



Figure 6: Exaggeration ratio for a true IC of 0.03 reported only when significant.

**What the results do not show.** None of this means published anomalies are false; Chen (2021) argues that the largest published $t$-statistics cannot be produced by $p$-hacking alone. What the results do show is that a cross-sectional IC is much less precise than the size of the panel suggests, that the shortfall can be calculated in advance, and that a small IC from a short sample should be treated as underpowered until shown otherwise.

## 10 Limitations

**The universe is completely survivorship-biased.** This is the most serious limitation and it cannot be repaired with the data available. All 2,559 tickers survive to the end, because the stock list is the index's membership on the download date; a point-in-time universe would have several times as many names and thousands of delistings, and the bias is not constant over time. My mean ICs are therefore not credible estimates of any economic quantity, and I do not present them as such. The paper's claims concern $\sigma_{IC}$ and $N_{\text{eff}}$, which are second moments and much less exposed. The cohort test in §8 supports this without proving it. A fully point-in-time panel could give a different $\sigma_F$, most plausibly a larger one, since delisting returns are volatile and cross-sectionally clustered in exactly the periods when factor realisations are extreme.

**One market, one sample period, two signals.** Everything is US equities, 2000-02–2026-02, price-based signals. I have not shown that $\sigma_F \approx 0.1397$ generalises to other markets, longer histories, or accounting-based signals. The *mechanism* should generalise wherever a signal correlates with factor loadings, but the magnitude is an empirical quantity and I measured it once.

**The simulation's IC is thin-tailed; real ICs are not.** As noted after Proposition 1, $g(f)$ is bounded, so the simulated IC has lighter tails than a normal. The realised series have kurtosis about 3.82, slightly heavy. The coverage figures in Table 3 are therefore mildly optimistic relative to real data, and inference on real IC series is likely to be slightly worse than the simulation suggests.

**The model is simple.** One factor, constant loadings, cross-sectionally uncorrelated residuals. Residual correlation from industry clustering would add a further non-diversifiable term, so relaxing (A3) worsens the conclusion. Time variation in $q$ or $\lambda$ likewise adds variance. I am not aware of a plausible relaxation that makes precision better.

**The snapshot is not exactly reproducible.** Adjusted closes drift as dividends accumulate, so a fresh download from Yahoo Finance should not be expected to match this file exactly. Returns computed from adjusted closes are still correct returns, so this limits exact reproducibility but does not bias the results, which are stated against my original download.

## 11 Conclusion

Past a modest number of stocks, the precision of a cross-sectional information coefficient depends on the number of periods and hardly at all on the number of stocks. Any usable signal is correlated with factor loadings, so each period's IC absorbs the realised factor return, one draw common to every name, and adding stocks does not average it away. The per-period variance is $\sigma_F^2 + \kappa/N$ with $\kappa \approx 1$, effective breadth is capped at $1/\sigma_F^2$, and the required sample length follows directly.

Measured on 2,559 US stocks over twenty-five years, $\sigma_F = 0.1397$, so a cross-section of 1,552 names is worth about 50 independent observations and the whole panel is worth 15,078 observations, not 465,750. Detecting a true IC of 0.03 needs 14.5 years for a single test and 27.2 years at the hurdle the field has adopted, and no ten-year window can detect it at all. My own momentum series gives 0.0123 with $t = 1.51$, which I report as a null.

The per-period floor is not new; it is strategy risk (Qian and Hua, 2004) and it bounds the attainable information ratio (Ding and Martin, 2017). What this paper adds is reading it as

a statement about statistical power: effective sample size, required $T$, and the exaggeration of published magnitudes. The calculation is cheap and can be done before a study is run.

**Reproducibility.** Code for every result in this paper is at `https://github.com/thotahimalnaidu/ic-power-analysis`. The price data come from Yahoo Finance and are not redistributed; `make download` fetches a fresh copy with `yfinance`, which will differ slightly from mine for the reason given in §10.

**Use of AI tools.** I used an AI assistant (Claude, Anthropic) for initial planning, help with debugging, language and grammar editing, presentation, and a final check of the manuscript. The research question, derivation, data analysis and conclusions are my own. I take full responsibility for the content and any errors.

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# A Proof of Proposition 1

**Conditional on the factor draw, the cross-section is i.i.d.** Fix $f \equiv f_{t+1}$. By (A6), each asset's $(s_i, \beta_i, \varepsilon_i)$ is an independent draw from the same joint normal distribution, and $r_i = \lambda s_i + \beta_i f + \varepsilon_i$ is linear in that draw. So conditional on $f$ the pairs $(s_i, r_i)_{i=1}^N$ are an i.i.d. sample from a bivariate normal distribution with correlation

$$\operatorname{corr}(s, r \mid f) \;=\; \frac{\lambda + qf}{\sqrt{\sigma_\varepsilon^2 + v_\beta f^2 + \lambda^2 + 2\lambda q f}} \;=\; g(f).$$

All of the dependence across assets runs through $f$, so once $f$ is fixed the *classical* formula (4) applies as it stands:

$$\operatorname{Var}(\widehat{IC}_t \mid f) \;=\; \frac{\left(1-g^2\right)^2}{N} + O(N^{-3/2}), \qquad \mathrm{E}[\widehat{IC}_t \mid f] \;=\; g - \frac{g\left(1-g^2\right)}{2N} + O(N^{-3/2}), \tag{19}$$

**Law of total variance.**

$$\begin{aligned}\operatorname{Var}(\widehat{IC}_t) &= \operatorname{Var}_f\Big[\mathrm{E}(\widehat{IC}_t \mid f)\Big] + \mathrm{E}_f\Big[\operatorname{Var}(\widehat{IC}_t \mid f)\Big] \\ &= \underbrace{\operatorname{Var}_f[g]}_{=\sigma_F^2} \;-\; \frac{1}{N}\operatorname{Cov}_f\!\big(g,\, g\left(1-g^2\right)\big) \;+\; \frac{1}{N}\,\mathrm{E}_f\Big[\left(1-g^2\right)^2\Big] \;+\; O(N^{-3/2}).\end{aligned}$$

The first term is (11) and does not involve $N$; the other two vanish as $N \to \infty$ and together are $\kappa/N$ as defined in (10). The covariance term comes from the conditional mean, which carries an $O(N^{-1})$ bias, and because that bias moves with $f$ it contributes to the variance at order $1/N$. Writing $\rho = \mathrm{E}_f[g]$ and using $\mathrm{E}_f[g^2] = \rho^2 + \sigma_F^2$, both terms expand to give $\kappa = 1 - 2\rho^2 - 3\sigma_F^2 + O(g^4)$, which is 0.972 in the calibration of §6. This gives (9). □

**Remark.** A derivation that tracks only the idiosyncratic term $u_t$ in (5) gets $\sigma_\varepsilon^2/(ND(f)^2)$ for the conditional variance, where $D(f)$ is the denominator of $g$ in (11). It misses the sampling error in $\operatorname{cov}_i(s, \beta)$ and in $\operatorname{sd}_i(r)$, which are of the same order. In the calibration of §6 that route understates the coefficient on $1/N$ by about 5%. The simulation estimates this coefficient from the data, which is how the error showed up. The $O(N^{-3/2})$ residual shows up in Table 1 as the slightly larger errors at small $N$.

## B The multi-factor statement

With $K$ factors, $r_{i,t+1} = \lambda s_{i,t} + \boldsymbol{\beta}_i' \mathbf{f}_{t+1} + \varepsilon_{i,t+1}$ and $\mathbf{q} = \mathrm{cov}_i(s, \boldsymbol{\beta}) \in \mathbb{R}^K$, equation (5) becomes $\mathrm{cov}_i(s, r) = \lambda + \mathbf{q}'\mathbf{f}_{t+1} + u_t$ and the floor is

$$\sigma_F^2 = \mathrm{Var}_{\mathbf{f}}\left[\frac{\lambda + \mathbf{q}'\mathbf{f}}{\sqrt{\sigma_\varepsilon^2 + \mathbf{f}' V_\beta \mathbf{f} + \lambda^2 + 2\lambda \mathbf{q}'\mathbf{f}}}\right], \qquad V_\beta = \mathrm{cov}_i(\boldsymbol{\beta}, \boldsymbol{\beta}').$$

To leading order in small $\mathbf{f}$ this is $\mathbf{q}'\Sigma_{\mathbf{f}}\mathbf{q}/\sigma_\varepsilon^2$. The floor is positive whenever the signal loads on any priced or unpriced common factor, and it does not depend on $N$.

## C Data checks

The script `src/audit.py` checks the raw files directly. All 2,559 tickers in the price file have data on its final date, 2026-03-11, so there are no delistings. In the fundamentals file, `book_to_market` is 83.2% negative with median $-0.805$, and its exponential has median 0.447, which is what log book-to-market looks like. `div_yield` takes only the values 0 to 3, and 69% of them are zero. Every close in the price file lies within that day's high-low range, allowing for floating-point rounding.